SMART ENERGY METER

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INTRODUCTION

The traditional paradigm of electricity metering, characterized by manual readings and limited data accessibility, is rapidly giving way to a new era of intelligent, connected devices. Smart power meters represent a paradigm shift, offering a plethora of benefits ranging from real-time energy monitoring and consumption insights to enhanced efficiency, sustainability, and cost savings.

PROBLEM STATEMENT

- Current energy meter does not provide informations like individual power consumption and offers limited functionality.

OBJECTIVE

- To develop a smart energy meter which monitors and provides overall and individual power consumption analysis.

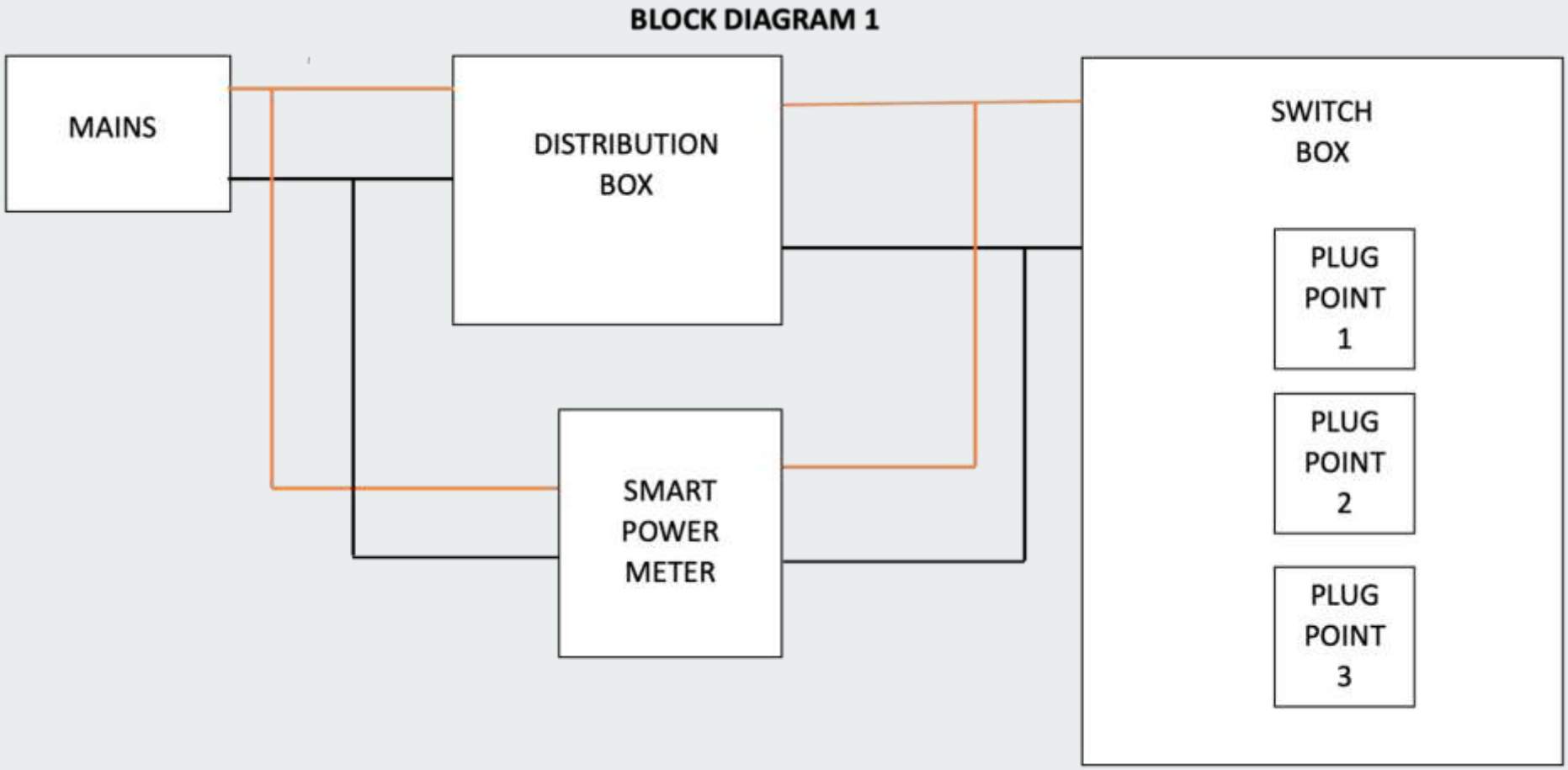


FIGURE 1: BLOCK DIAGRAM OF OVERALL SYSTEM

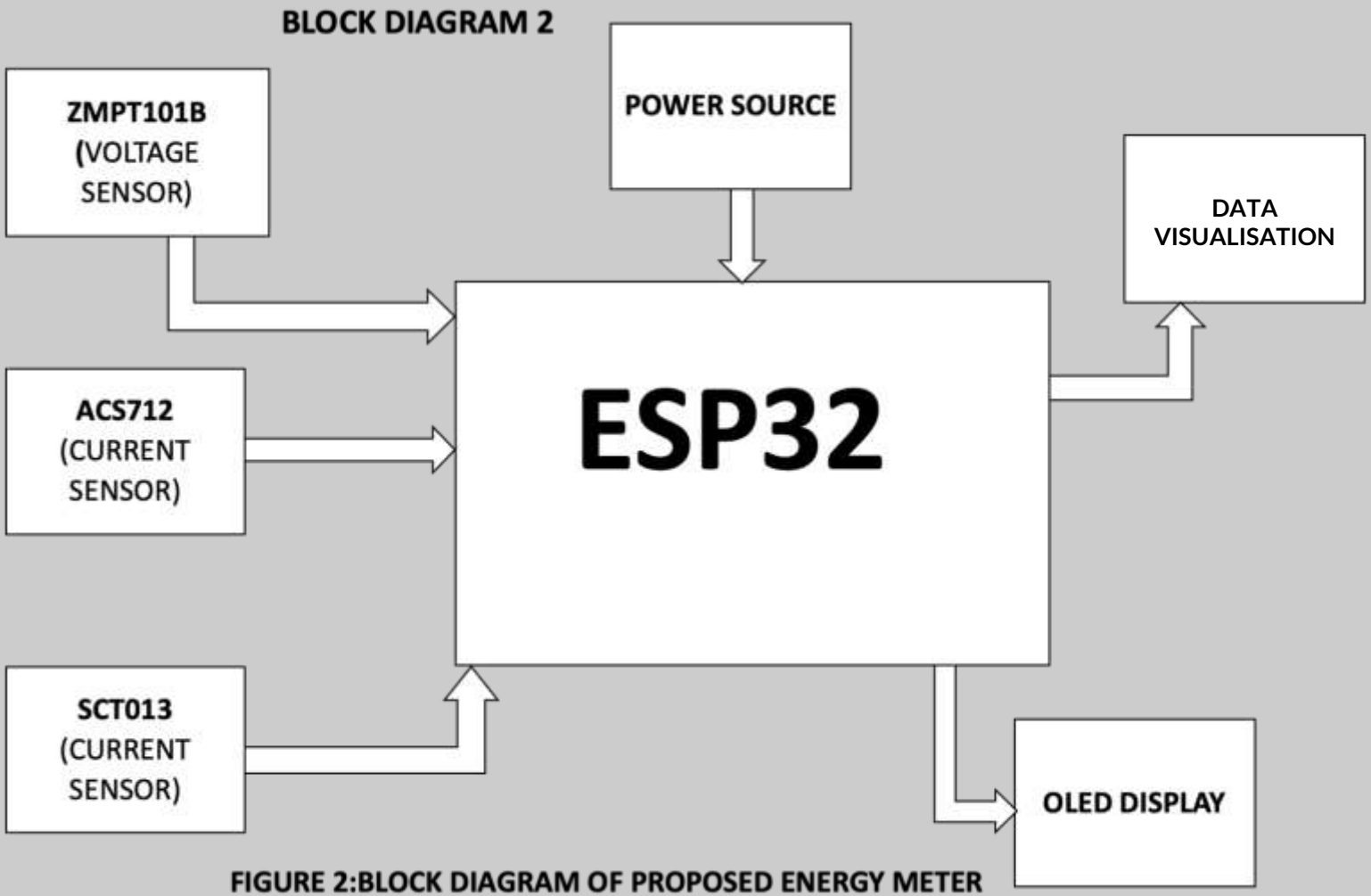


FIGURE 2:BLOCK DIAGRAM OF PROPOSED ENERGY METER

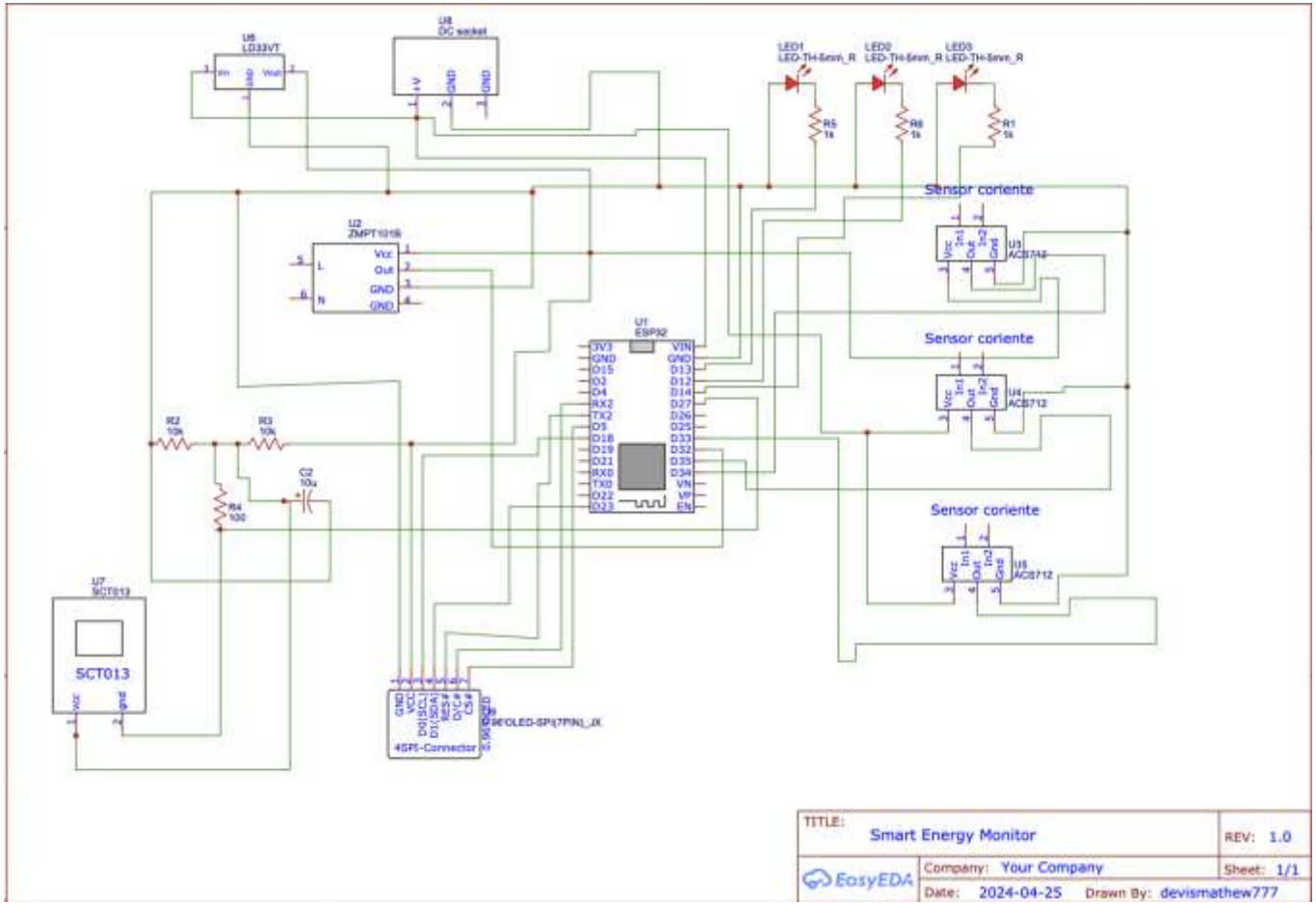


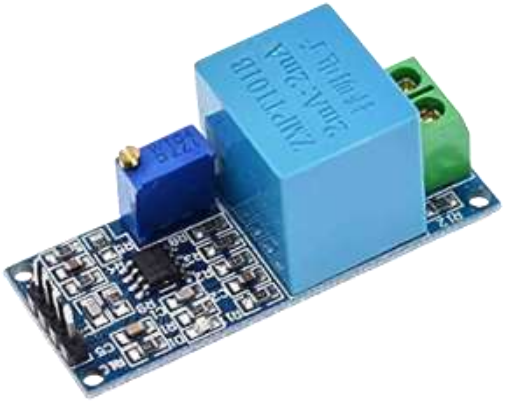
FIG 3: SCHEMATICS OF SMART POWER METER

HARDWARE DETAILS

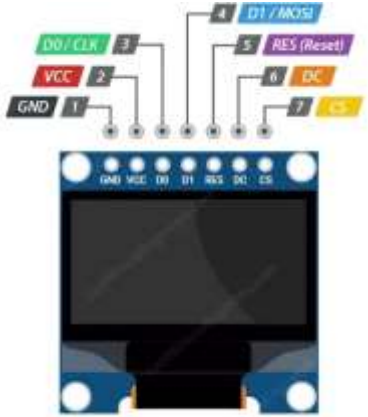
The **ACS712** is a current sensor,with a maximum detection range of 30A.

The **SCT013** is a current sensor module with maximum detection range of 100A

HARDWARE DETAILS

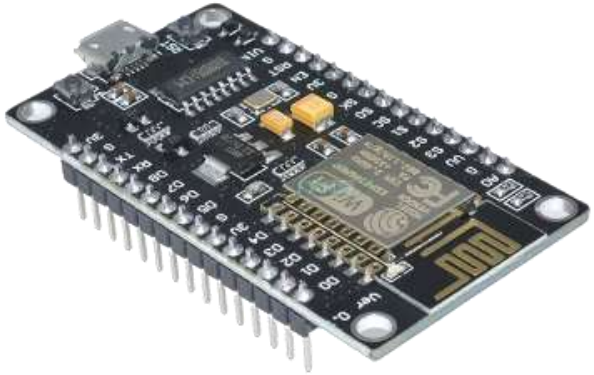


The **ZMPT101B** is a voltage sensor principle with a maximum range of 250V



OLED display 128x64 SPI is a compact and versatile module, ideal for applications requiring crisp visuals in limited spaces.

HARDWARE DETAILS



ESP32 devkit v1 is a microcontroller board.It boasts a dual-core Tensilica LX6 processor running at up to 240 MHz,. With built-in Wi-Fi and Bluetooth connectivity, it enables seamless communication and networking capabilities for IoT applications

SOFTWARE DETAILS

Firebase, a platform developed by Google, offers tools and services for real-time databases to authentication, cloud storage, hosting, analytics, and more.

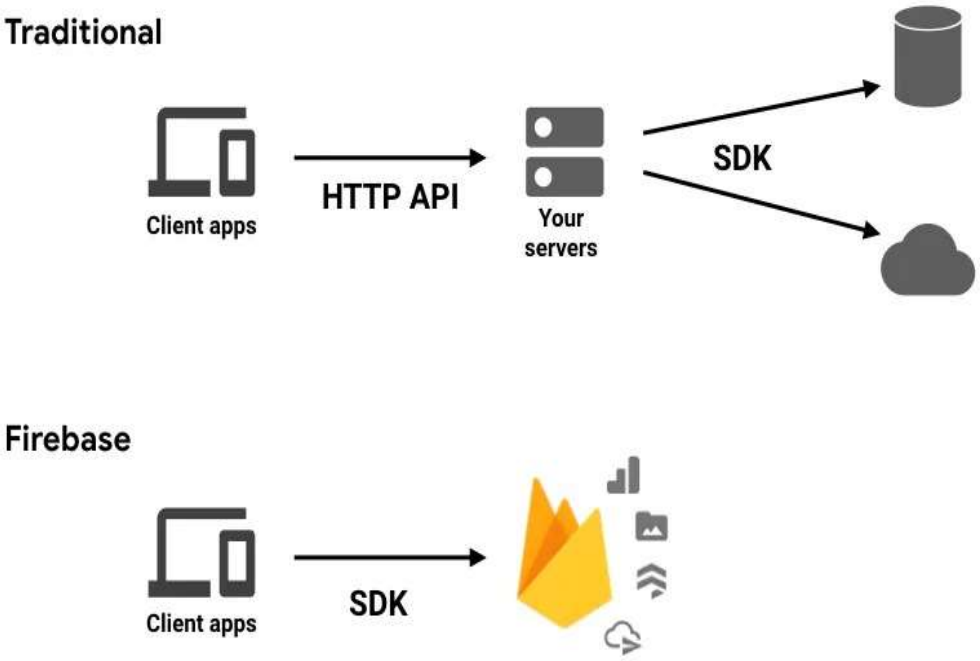


FIG 4: FIREBASE

SOFTWARE DETAILS

PlatformIO simplifies programming for **esp32 devkit v1** by providing a simple and convenient development environment.

Python IDE is a software tool designed to facilitate Python applications.

We utilised python for developing a frontend for our project.

SOFTWARE DETAILS

Google Sheets

Google Sheets is used for storing data in spreadsheet format.

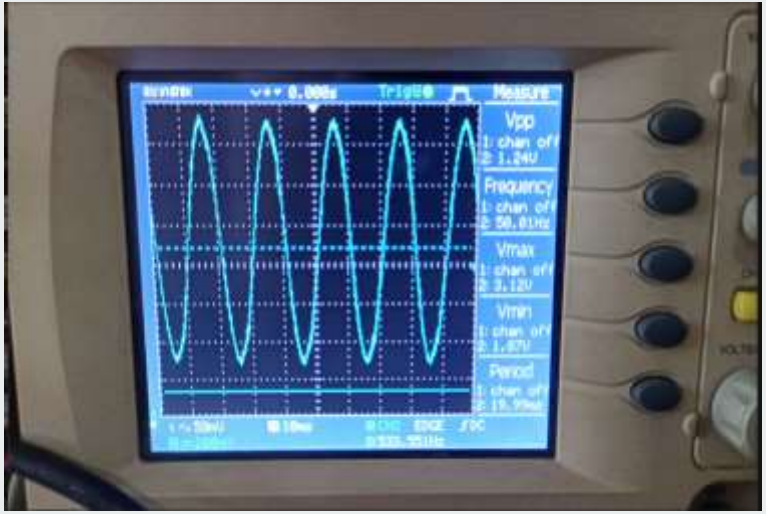
Here we utilized its built-in tools for data visualization, such as charts and graphs, to represent their data effectively.

WORK DIVISION

DATE	SCHEDULE
19 FEBRUARY	Idea presented
19 FEBRUARY 2024	Abstract approval
22 FEBRUARY 2024	PPT Presentation
24 FEBRUARY 2024	Designing Basic Idea
1 MARCH 2024	Start implementation
15 MARCH 2024	Stage 1 completion and testing
7 APRIL 2024	Stage 2 completion and testing
17 APRIL 2024	Stage 3 completion and testing
26 April 2024	Start assembling and testing
2 MAY 2024	Final Testing & completion

PHASE - 1

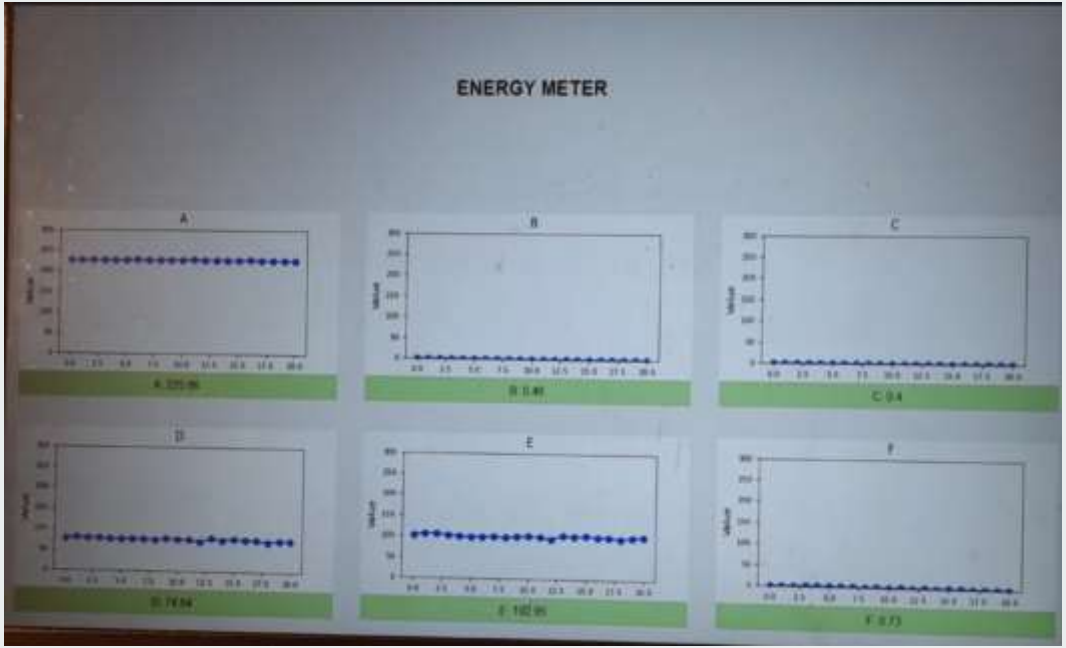
- We finalised the components to buy and decided the workflow .
- Components was acquired and started working on it.
- Started with calibrating the sensors using oscilloscope from the lab.
- Started working on the development of distribution board capable of handling a 5KW load

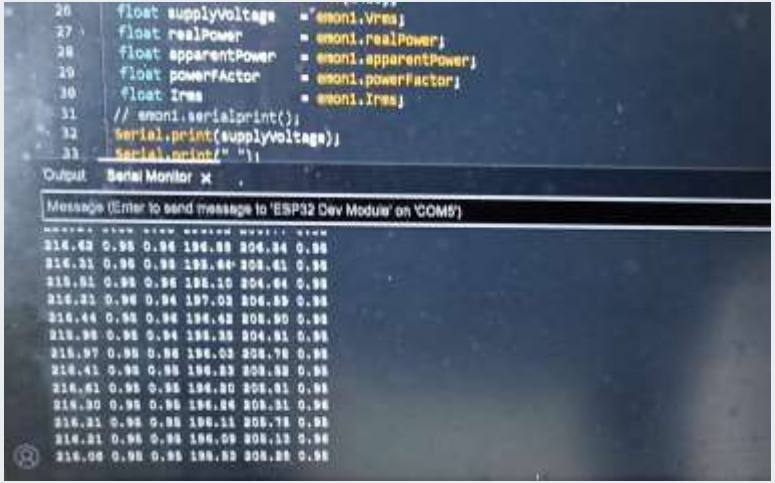


PHASE - 2

Started working on software side for programing the microcontroller and reading the values from the sensors and calculating the power and other parameters.

Used python programming for plotting the values received from microcontroller.

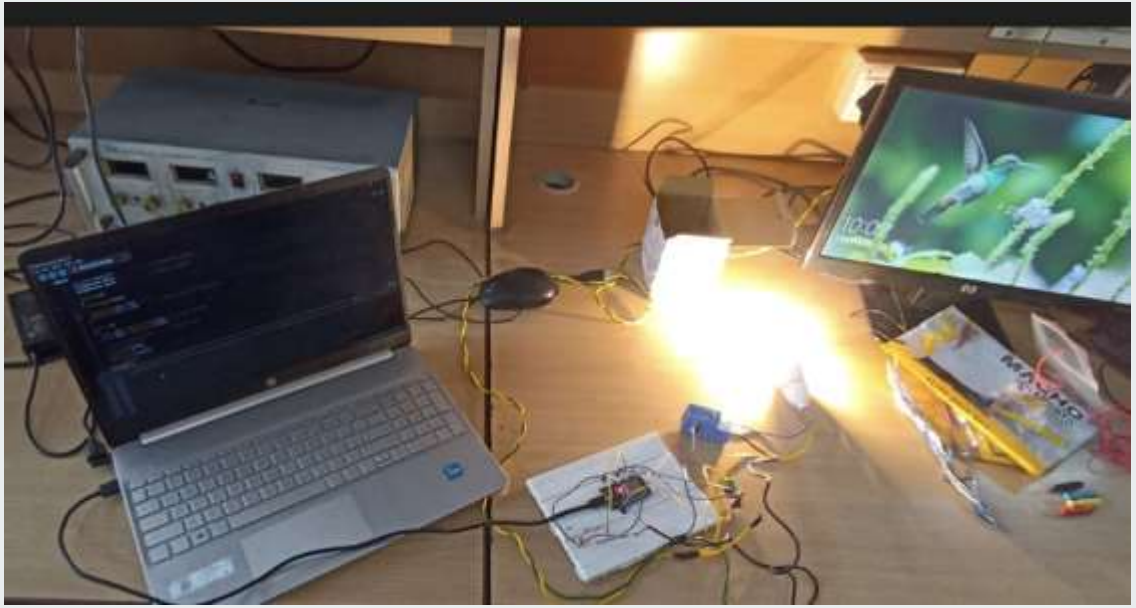




PHASE - 3

Successfully Implemented the program to use a 100W incandescent light bulb to read the electrical parameters.

Tested it on a 5K load from the Electrical Lab.



OUTPUT

voltage_expected(in V)	voltage_recorded(in V)	total_load Current_expected(in A)	total_load Current_recorded(in A)	apparent_power_expected()	apparent_power_rec
224	228	2.09	2.1	468.16	483
220	226	4.21	4.22	926.2	976
223	227	6.3	6.28	1404.9	1443
221	227	8.5	8.27	1878.5	1869
221	227	10.5	10.27	2320.5	2353
216	214	13	12.23	2808	2638
217	214	15.18	14.18	3294.06	3069
217	213	17.39	16.59	3773.63	3579
215	215	19.46	18.89	4183.9	4069
216	216	21.57	21.32	4659.12	4567
215	214	23.85	22.63	5127.75	5032

FIG 5: VALUES

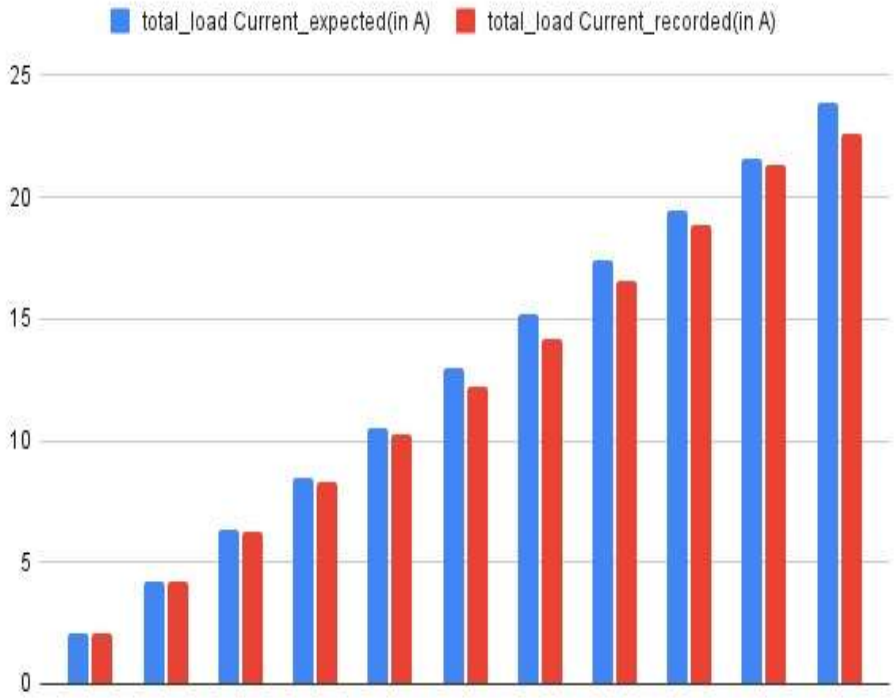


FIG 6: Total load current expected and recorded

OUTPUT

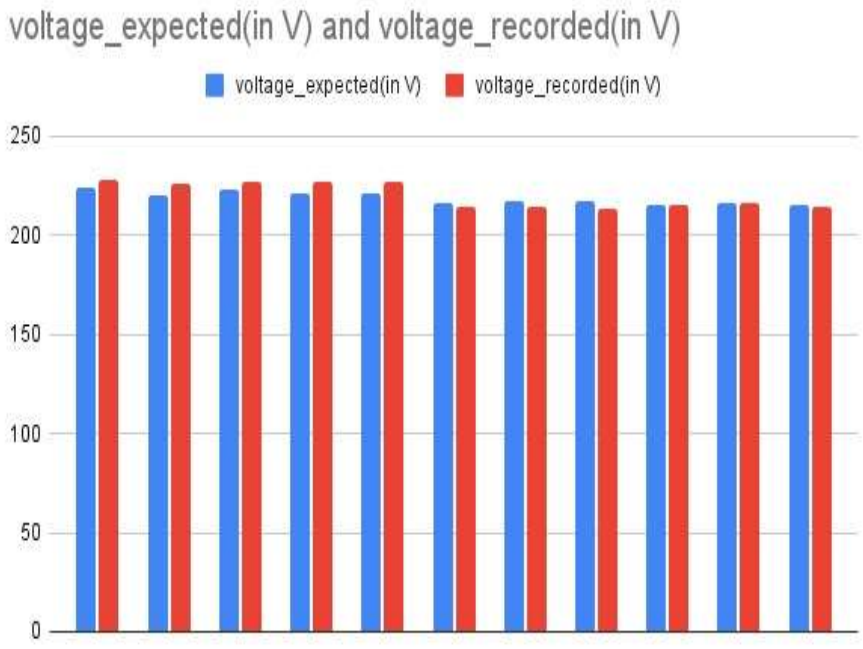


FIG 7: Voltage expected and recorded

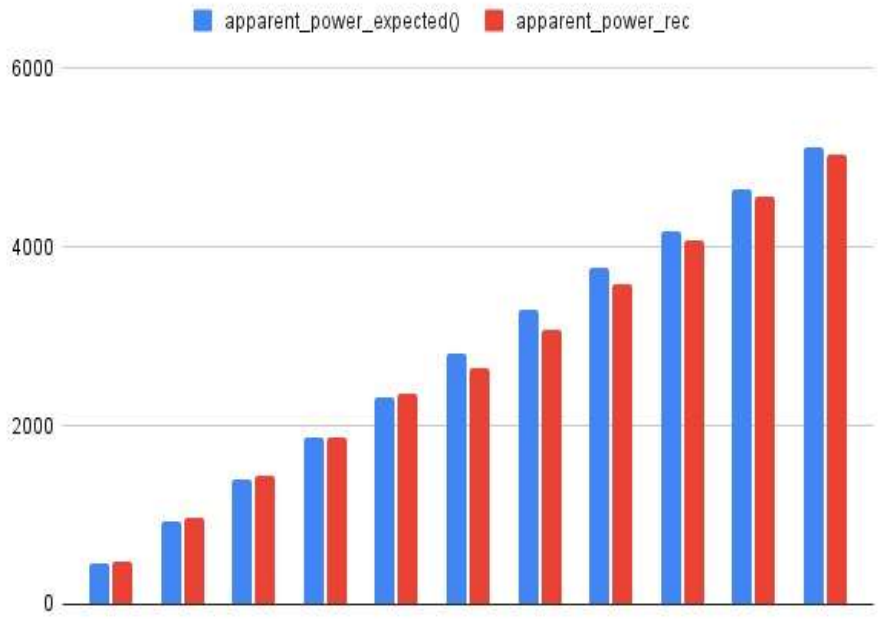


FIG 8: Apparent power expected and recorded

RESULTS

- We have successfully developed an energy meter capable of **single-phase energy monitoring**. The meter provides **real-time data showing the power consumption in each room.**

- We've encountered challenges with accurately measuring small currents

1. **Usage of ADC channel 2 of esp32 is limited while enabling wifi module**
2. **Current sensors are not able to accurately sense value below 1A**
3. **The ESP32 necessitates additional time to yield stable output readings**

FUTURE SCOPE

Three-Phase Monitoring:

1. Hardware modifications to support three-phase energy monitoring.
2. Development of algorithms for calculating total power consumption, power factor, and other relevant metrics specific to three-phase systems.
3. Update of the user interface to effectively display three-phase energy data.

Implementation of AI for Enhanced Analysis:

1. Integration of AI algorithms, such as machine learning, for advanced data analysis.
2. Analysis of energy usage patterns.
3. Provision of insights to optimize energy efficiency in three-phase systems.

FUTURE SCOPE(PRODUCT DESIGN)

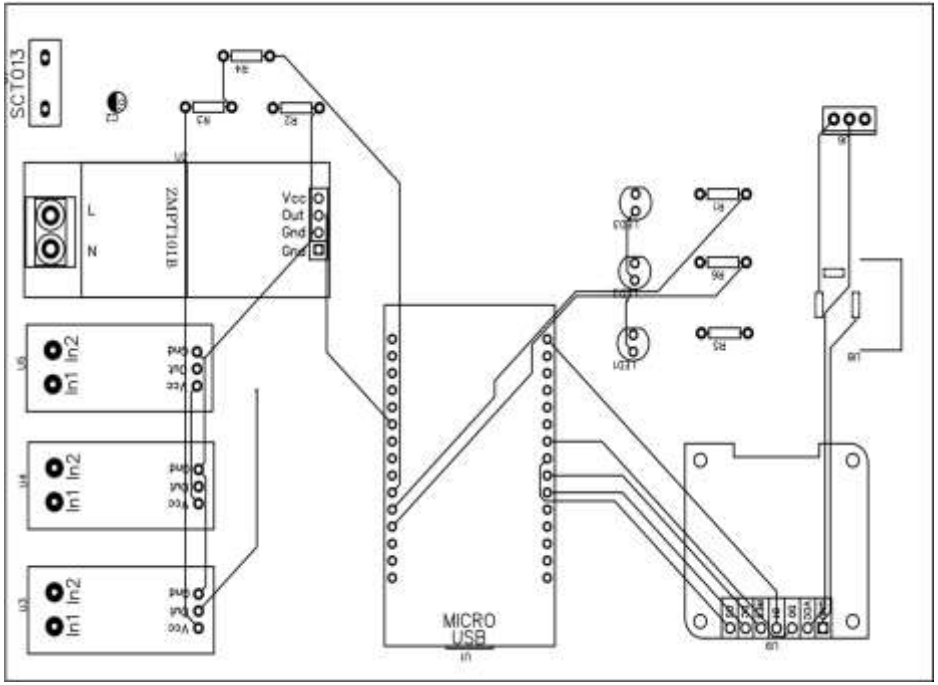


FIG 9:BOTTOM LAYER

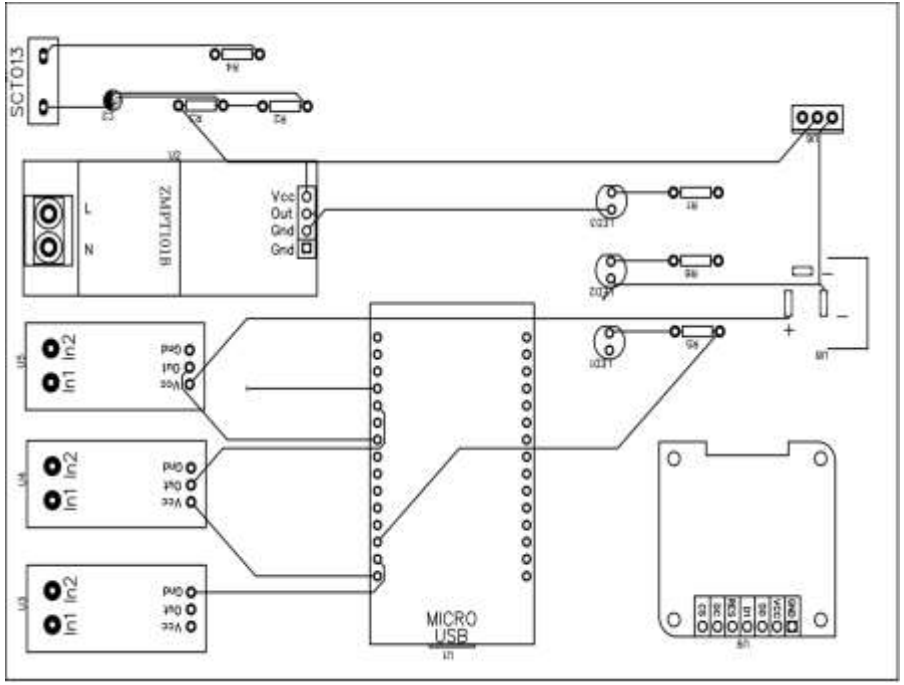


FIG 10:TOP LAYER

FUTURE SCOPE(PCB PROTOTYPE)

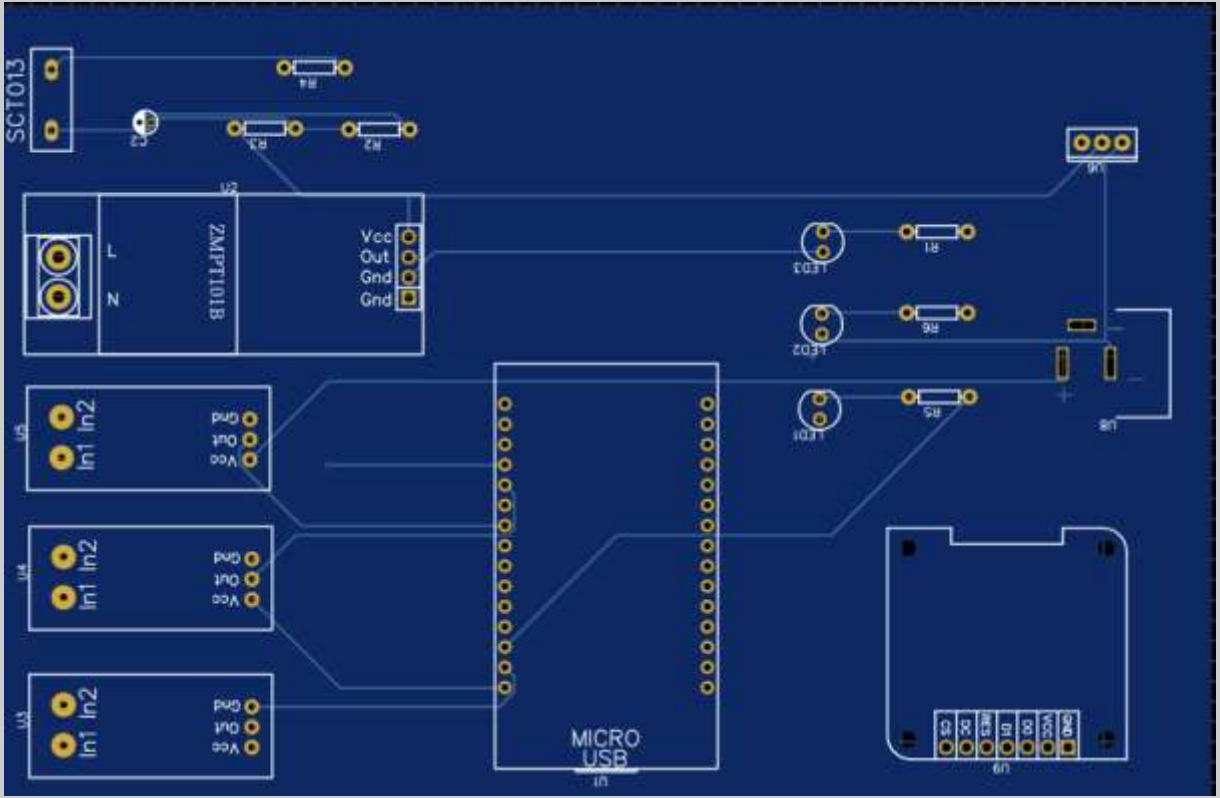


FIG 11: PROTOTYPE

CONCLUSION

We have successfully developed a smart energy meter which monitors and provides overall and individual power consumption analysis.

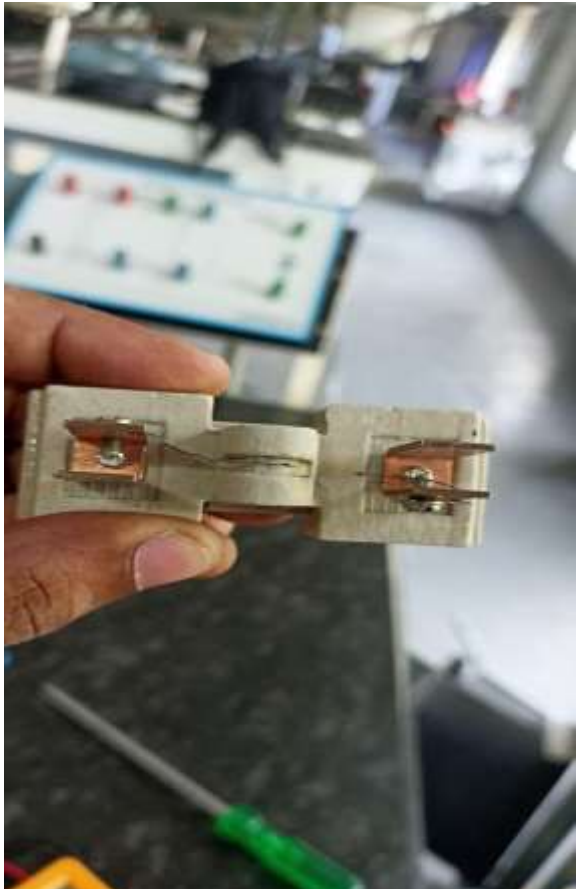
REFERENCE

- Macheso, Paul Stone, and Doreen Thotho. "ESP32 Based Electric Energy Consumption Meter." International Journal of Computer Communication and Informatics 4, no. 1 (2022): 23-35.

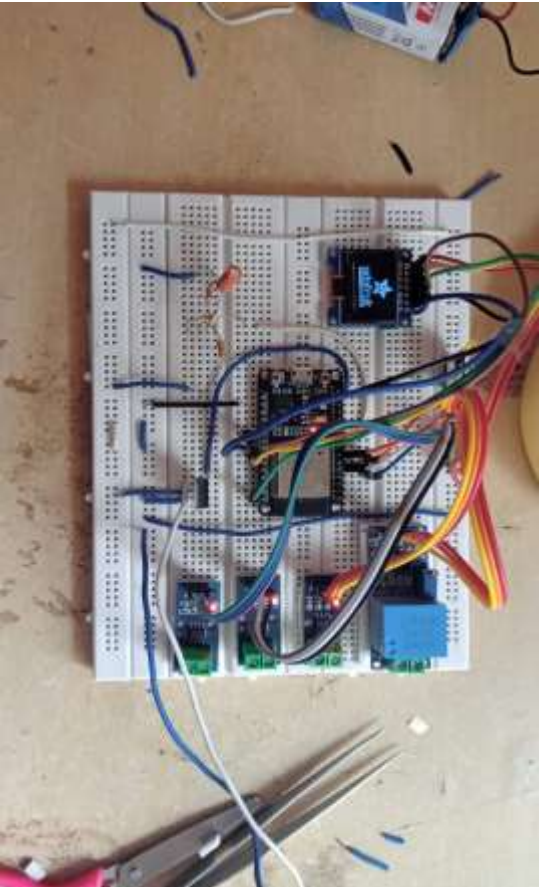
- Navlani, Avinash, Armando Fandango, and Ivan Idris. Python Data Analysis: Perform data collection, data processing, wrangling, visualization, and model building using Python. Packt Publishing Ltd, 2021.

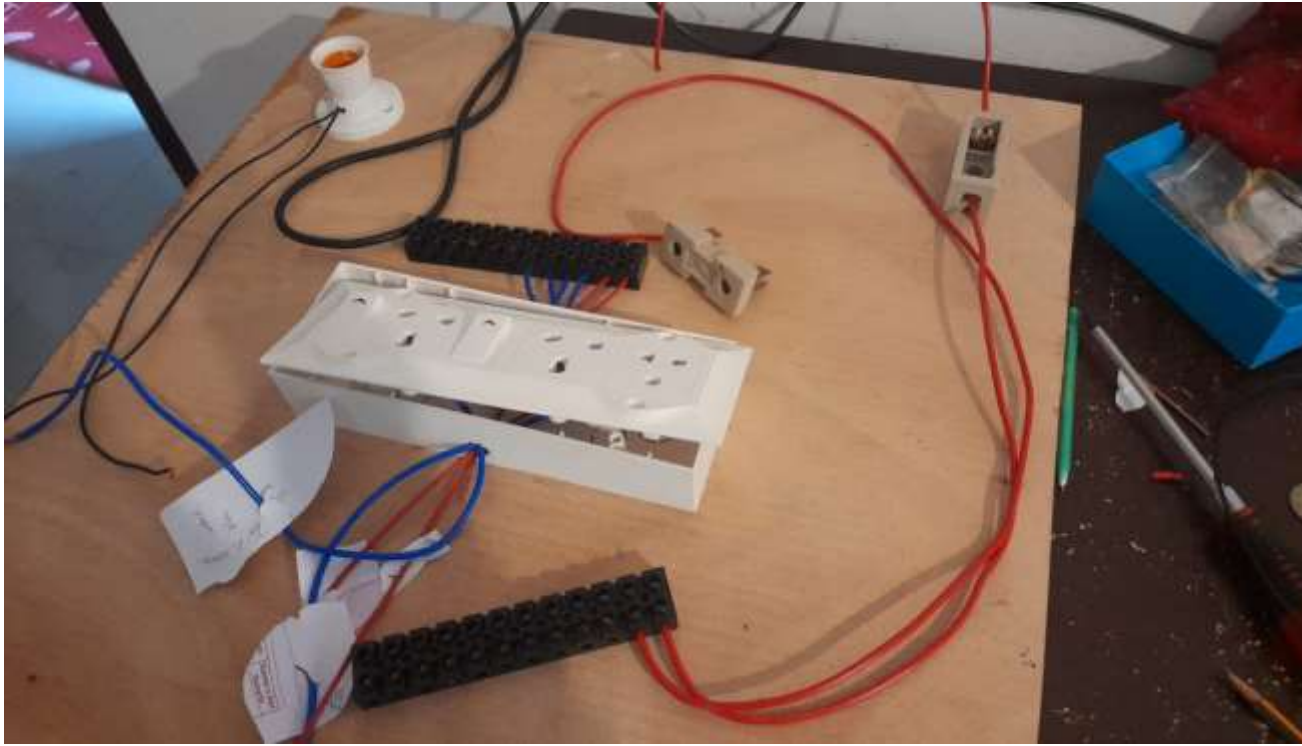
- Preethi, V., and G. Harish. "Design and implementation of smart energy meter." In 2016 International Conference on Inventive Computation Technologies (ICICT), vol. 1, pp. 1-5. IEEE, 2016.

Project gallery

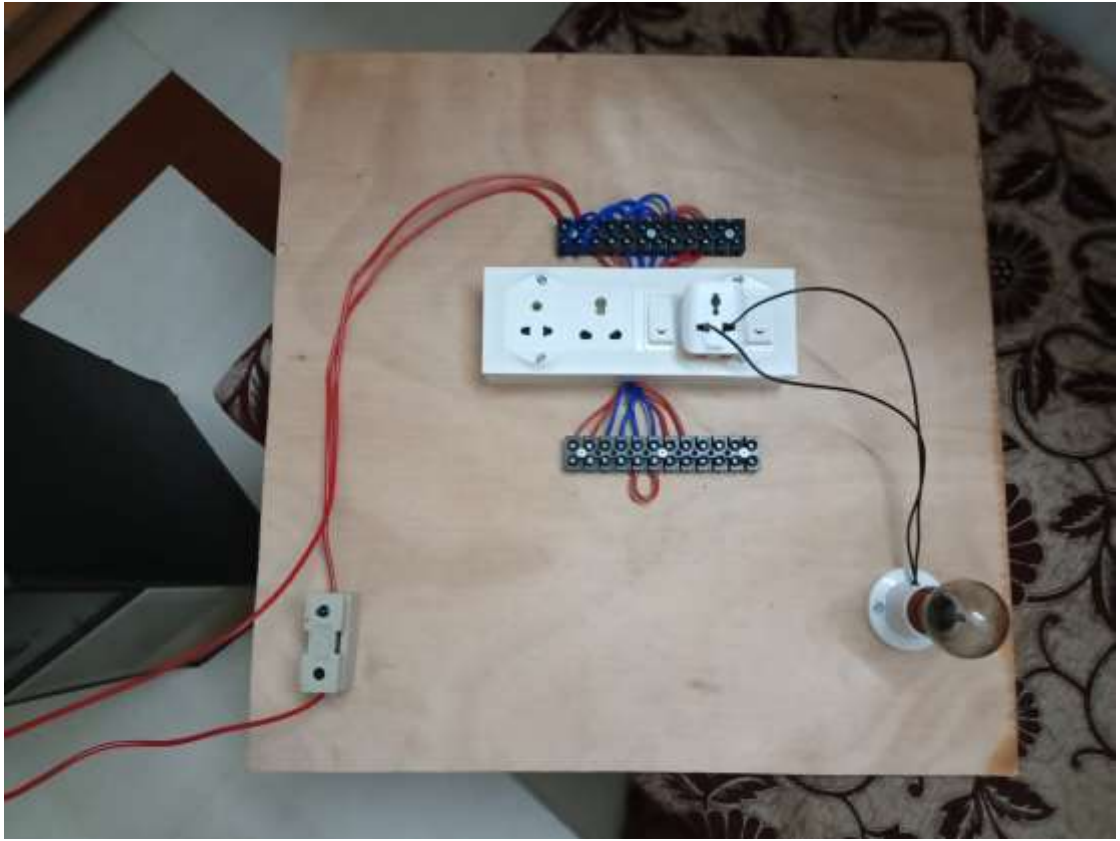


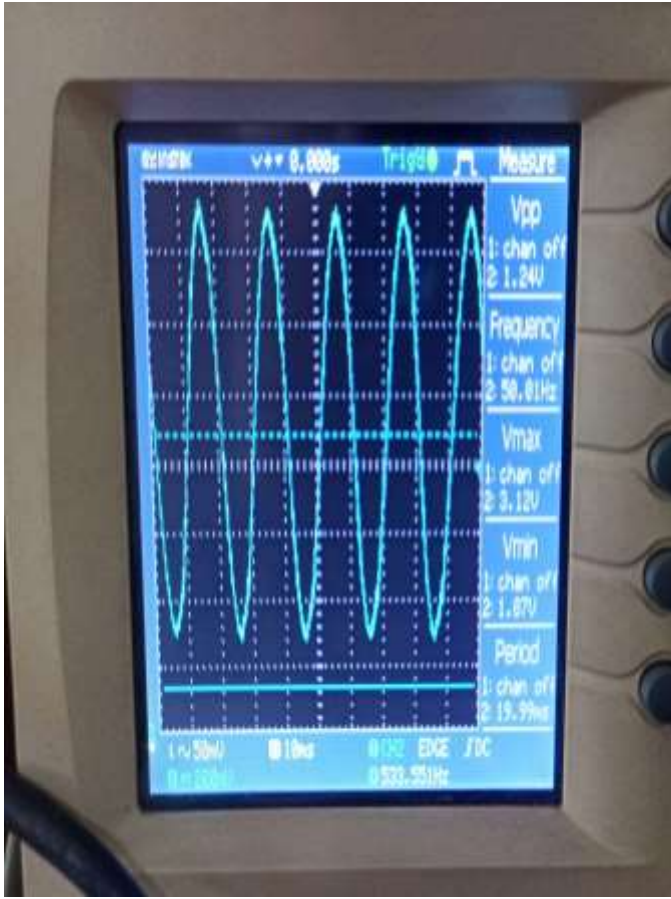
Project gallery





Project gallery





THANK YOU